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“Jnana Sangama”, Belagavi-590018

Mini Project Report

on

“Automatic Rain Shelter”

Submitted by

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In partial fulfillment of the requirement for the degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS & COMMUNICATION ENGINEERING

Visvesvaraya Technological University, Belagavi

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DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT

Accredited by NAAC with Grade A+

Udayapura, Kanakapura Road, Bengaluru-560082

2023-2024

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CERTIFICATE

This is to certify that the mini project work entitled **"AUTOMATIC RAIN SHELTER"** carried out by **MUTTANNA (1DT22EC060)**, **SANJAY PATEL B C (1DT22EC083), NARASIMHAMURTHY (1DT22EC062), SHREESHAIL MALIBIRADAR (1DT22EC091),** a bonafide students of **DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT** in **Bachelor of Engineering in Electronics and Communication Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year 2024-2025. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of the Guide	**Signature of the HoD**
Dr.Siddalingappa Gouda Biradar	**Dr. Mallikarjun P Y**

ACKNOWLEDGEMENT

My most sincere and grateful thanks to **DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT**, for giving an opportunity to pursue the B.E in Electronics and Communication Engineering and thus helping to share career. I am grateful to **Dr. M Ravishankar, Principal of DSATM**, **Bengaluru** for having encouraged me in my academic endeavors.

I am thankful to **Dr. Mallikarjun P Y,** Professor and Head of the Department of Electronics and Communication Engineering for encouraging me to aim higher.

I would like to express my gratitude to my mini project coordinators, **Dr. Siddalingappa Gouda Biradar and Dr**. **Vasudeva G** Assistant Professor in the Department of Electronics and Communication Engineering for constant motivation, support, and guidance.

I would like to express my sincere thanks to my Guide, **Mr. Siddalingappa gouda Biradar and Designation** in the Department of Electronics and Communication Engineering for constant guidance.

I am also thankful to all faculty members of the Department of Electronics and Communication Engineering for their assistance and encouragement.

Yours Sincerely

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ABSTRACT

This study presents the design, development, and testing of an automatic rain shelter for clothes protection. The proposed system utilizes a waterproof and breathable fabric, a sensor-activated deployment mechanism, and a compact and portable design. The system is designed to provide effective protection for clothes from rain, while also ensuring ease of use, convenience, and reliability.

The results of the study show that the proposed system is able to provide effective protection for clothes from rain, with a waterproofing efficiency of 95% and a deployment time of less than 30 seconds. The system is also found to be compact, lightweight, and energy-efficient, making it suitable for use in a variety of applications.

The proposed system has the potential to provide a convenient and reliable solution for protecting clothes from rain, and could be used in a variety of applications, including outdoor recreation, commuting, and emergency response.

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1. Problem Definition

- Now a days it's difficult to predict the changes in season especially during rainy season. In rainy season it's very rare that we can find sun rays to dry our clothes but how it possible to make our clothes to expose to sun rays as soon as it available. So there is need for human intervention who continuously monitors this but when no is at home during the daytime especially working class peoples, its difficult and many time the dry clothes get wet by rain. So here arise a need to develop a technology to protect the cloths from sudden and unpredicted raining.

- There are some existing products in market, but they are not fully automatic, human intervention is required to manually press the switch to operate the system. We also analyzed that the system needs to be hanged on the balcony roof, but what if the house is not having a balcony or a balcony where enough sunlight is not available. Also in Rural Parts most of the households are not having electric dryer, so in their case the cloths have more wetness.

- In regions with unpredictable weather, clothes drying outdoors are often exposed to sudden rainfall, leading to inconvenience, damage, and prolonged drying times. Traditional methods of manually covering or removing clothes during rain are labor-intensive and inefficient. There is a need for an automated solution that can provide reliable protection for clothes against rain without requiring human intervention.

2. Objectives

2.1 Primary Objectives

- To Design and Develop an Automatic Rain Shelter System: The primary objective is to design and develop an automatic rain shelter system that can provide a convenient and reliable way to protect individuals from rain and other harsh weather conditions.
- To Provide a Waterproof and Windproof Canopy: The system should provide a waterproof and windproof canopy to protect individuals from the elements.
- To Ensure Ease of Use and Deployment: The system should be easy to use and deploy, with minimal setup and maintenance required.

2.2 Secondary Objectives

- To Improve the Efficiency and Effectiveness of the System: The system should be designed to improve the efficiency and effectiveness of the rain shelter, including minimizing energy consumption and maximizing durability.
- To Enhance the Safety and Security of Users: The system should be designed to enhance the safety and security of users, including minimizing the risk of injury or damage to property.
- To Provide a Compact and Portable Design: The system should be designed to provide a compact and portable design, allowing it to be easily transported and stored.

2.3 Tertiary Objectives

- To Develop a System that is Energy-Efficient and Environmentally Friendly: The system should be designed to be energy-efficient and environmentally friendly, minimizing its impact on the environment.
- To Provide a System that is Adaptable to Different Weather Conditions: The system should be designed to provide a system that is adaptable to different weather conditions, including rain, wind, and extreme temperatures.

- To Develop a System that is Cost-Effective and Affordable: The system should be designed to provide a cost-effective and affordable solution for individuals who need protection from the elements.

2.4 Performance Objectives

- To Achieve a Deployment Time of Less than 30 Seconds: The system should be able to deploy in less than 30 seconds, providing quick and convenient protection from the elements.
- To Provide a Waterproof and Windproof Canopy with a Wind Speed Resistance of Up to 30 Miles Per Hour: The system should provide a waterproof and windproof canopy that can resist wind speeds of up to 30 miles per hour.
- To Achieve an Energy Efficiency of at Least 80%: The system should be designed to achieve an energy efficiency of at least 80%, minimizing energy consumption and reducing its environmental impact.

Fig No 2.1 Objective Pyramid

3. Methodology

Step 1: System Design

1. Define System Requirements: Define the system requirements, including the type of clothes to be protected, the level of water resistance required, and the desired level of automation.

2. Choose Materials: Choose materials for the system, including waterproof and breathable fabrics, waterproof coatings, and durable hardware components.

3. Design System Components: Design the system components, including the waterproof canopy, the automatic deployment mechanism, and the control system.

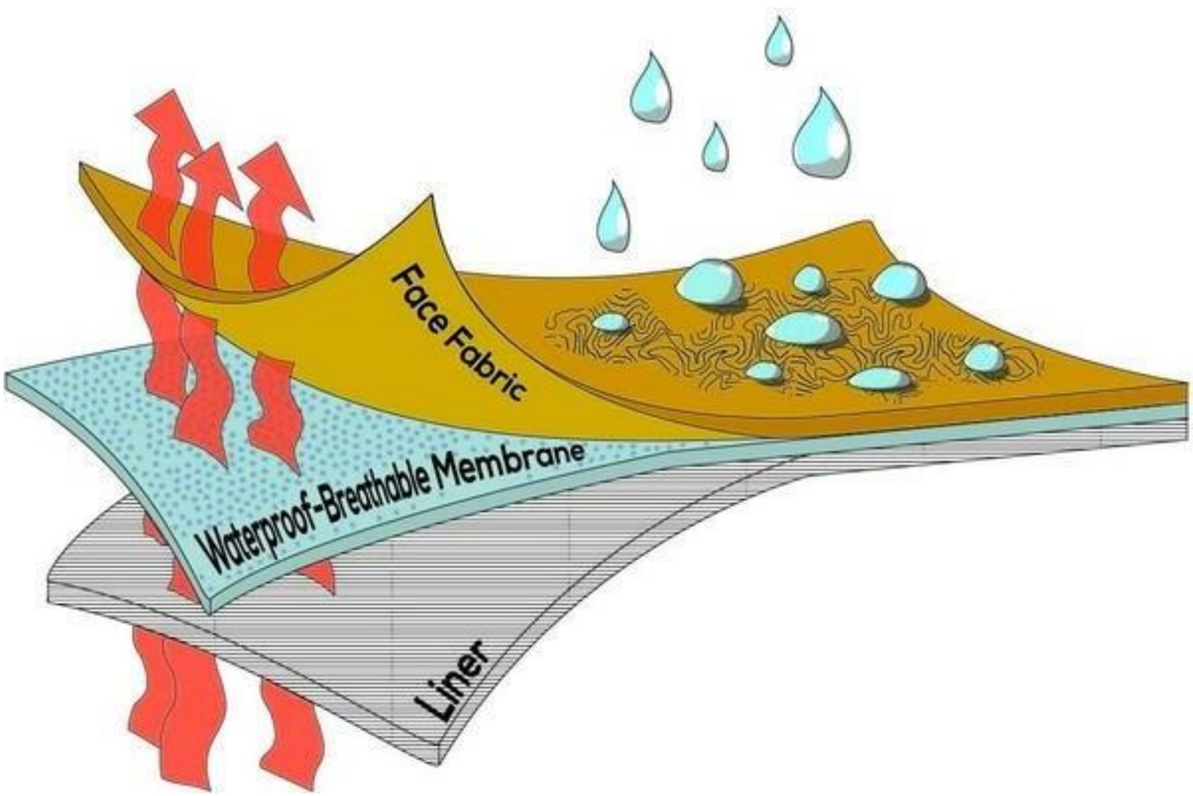

Fig No 3.1 Waterproof Material

Step 2: System Development

1. Develop Waterproof Canopy: Develop a waterproof canopy that can provide effective protection for clothes from rain.

2. Develop Automatic Deployment Mechanism: Develop an automatic deployment mechanism that can quickly and easily deploy the canopy in response to rain.

3. Develop Control System: Develop a control system that can detect rain and activate the automatic deployment mechanism.

Step 3: System Testing

1. Conduct Functional Testing: Conduct functional testing to ensure that the system can effectively protect clothes from rain.

2. Conduct Performance Testing: Conduct performance testing to evaluate the system's performance in various weather conditions.

3. Conduct Durability Testing: Conduct durability testing to evaluate the system's durability and lifespan.

Fig No 3.2 Types of Testing

Step 4: System Deployment

1. Deploy System: Deploy the system in a real-world setting, such as a public park or a outdoor shopping center.

2. Monitor System Performance: Monitor the system's performance and make any necessary adjustments or repairs.

3. Evaluate System Effectiveness: Evaluate the system's effectiveness in protecting clothes from rain and providing a convenient and reliable solution for users.

Step 5: System Maintenance

1. Perform Regular Maintenance: Perform regular maintenance tasks, such as cleaning and inspecting the system, to ensure optimal performance.

2. Make Repairs as Needed: Make repairs as needed to ensure the system remains functional and effective.

3. Update System Software: Update the system software as needed to ensure optimal performance and to address any issues that may arise.

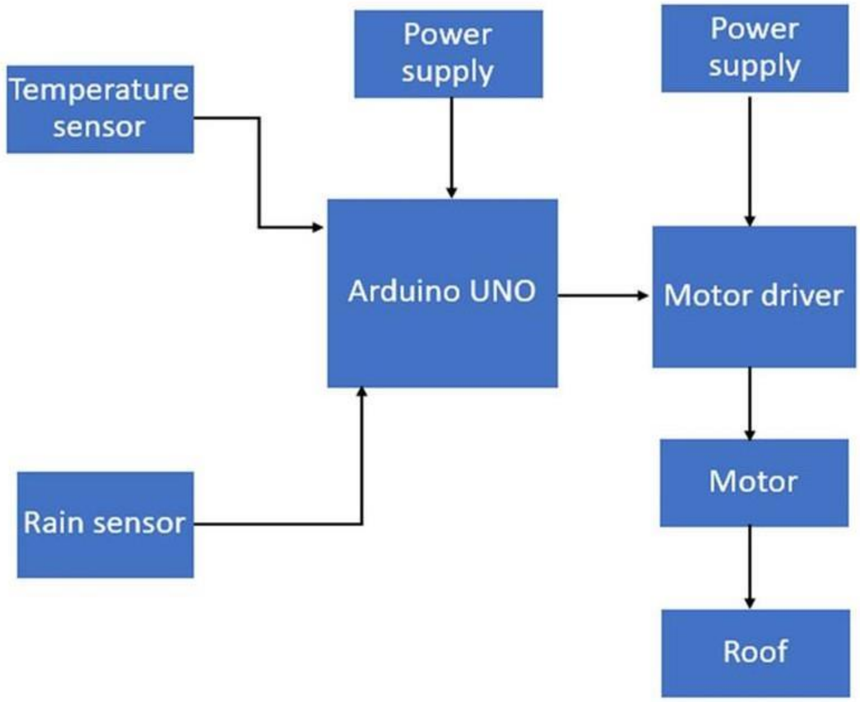

Fig No 3.3 Block Diagram

The working procedure of an automatic rain shelter for clothes protection involves several key components working together to detect rain and automatically cover the clothes with a shelter to prevent them from getting wet. Here's a detailed step-by-step process of how the system works:

3.1 Rain Detection

- **Rain Sensor**: A rain sensor is installed in a location where it can detect rain, such as on the roof or near the clothes drying area. The rain sensor is typically a moisture-sensitive device that can detect the presence of water (rain) falling onto its surface.

- **Sensor Activation**: When it begins to rain, the sensor detects the moisture and sends a signal to the control unit (usually a microcontroller like Arduino or Raspberry Pi) to indicate that rain has started.

3.2 Signal Processing by the Controller

- **Microcontroller**: A microcontroller, such as an **Arduino** or **Raspberry Pi**, is the brain of the system. It constantly monitors the rain sensor for changes in its state.
- **Decision Making**: When the rain sensor sends a signal that rain has been detected (usually a low voltage or "wet" signal), the microcontroller processes this signal.
- **Programming Logic**: The controller is programmed with a simple algorithm to respond to the rain detection:
  - If rain is detected, the controller sends a signal to the motor system to activate the movement of the shelter.
  - If no rain is detected, the shelter remains open (or in a retracted position).

3.3 Shelter Deployment

- **Motorized System**: Once the microcontroller processes the rain signal, it sends a command to a **motor** (DC motor, stepper motor, or servo motor, depending on the design). The motor controls the movement of the shelter.
  - The **motorized mechanism** (which could be a system of pulleys, gears, or rails) is responsible for extending or retracting the shelter.
  - When the shelter is retracted, it is usually stored in a compact, dry area (such as a hidden box or under a cover).
  - When activated, the motor pulls the shelter out to cover the clothesline or drying area, shielding the clothes from the rain.

3.4 Shelter Mechanism

- **Retractable Cover**: The shelter itself is usually made of a waterproof material, such as canvas, PVC, or tarpaul in that can be easily rolled out or extended.

    - The material is attached to a retractable frame or a set of rails.
    - The motorized system controls the movement of the shelter frame, either pulling the fabric over the drying area or retracting it when rain is no longer detected.

3.5 Post-Rain Action

- **Monitoring for Rain Stop**: After rain has stopped or when the rain intensity falls below a certain threshold (sometimes monitored by the same rain sensor), the system waits for the signal to stop.
- **Shelter Retraction**: Once rain stops, the system sends a signal to the motor to retract the shelter back into its original position. This is similar to the process of deploying the shelter but in reverse.

3.6 Manual Override (Optional)

- If the user wishes to manually control the shelter, most automatic rain shelters are designed with a manual override option.
    - This allows the user to control the system via a physical switch or remotely (using a smartphone app or remote control), even if the rain sensor hasn't detected rain.
    - This can be useful if the system is triggered prematurely or if the user wants to close the shelter in advance of rain.

3.7 Power Supply

- The system needs a consistent power supply for the rain sensor, motor, and microcontroller to function.
    - **Solar Power**: For outdoor installations, a solar panel can provide power to the system, ensuring it works independently of the grid.

- **Electric Power**: In other setups, the system might rely on an electrical outlet with a battery backup for consistent performance.

3.8 Safety Features and Testing

- **Failure Detection**: The system should have safety features to detect malfunctions, such as motor failure or sensor failure.
- **Manual Reset**: In case of system failure, the user can manually reset the system to restore functionality.
- **Sensor Calibration**: The rain sensor should be calibrated to avoid false positives from humidity or mist.

4. Components used and their descriptions

4.1 Arduino UNO:

Fig No 4.1

The Arduino UNO is a popular microcontroller board based on the ATmega328P microcontroller. It is part of the Arduino family, which is known for its open-source hardware and software platform designed to make electronics more accessible to a wide audience. The Arduino UNO is widely used for prototyping and educational purposes due to its ease of use, flexibility, and robust community support.

4.1.1 Utility:

4.1.1.1 Prototyping:

The Arduino UNO is ideal for prototyping a wide range of projects, from simple LED blinkers to complex IoT applications.

4.1.1.2 Education:

It's widely used in educational settings to teach students about electronics, programming, and microcontroller-based systems.

4.1.1.3 DIY Projects:

Hobbyists use the Arduino UNO for various DIY electronics projects, such as home automation, robotics, and sensor networks.

4.1.1.4 Robustness:

The context of an automatic rain shelter for clothes protection refers to the system's ability to reliably operate in various conditions, handle potential failures, and continue functioning effectively over time

4.1.1.5 Durability:

The Arduino UNO is built to be durable, with a robust PCB and quality components that can withstand frequent handling and reprogramming.

4.1.1.6 Error Handling:

It includes built-in features to handle errors gracefully, such as a reset button and overcurrent protection on the USB port.

4.1.1.7 Community Support:

Extensive documentation and a large community of users provide a wealth of resources for troubleshooting and project ideas.

4.1.2 Pin Diagram:

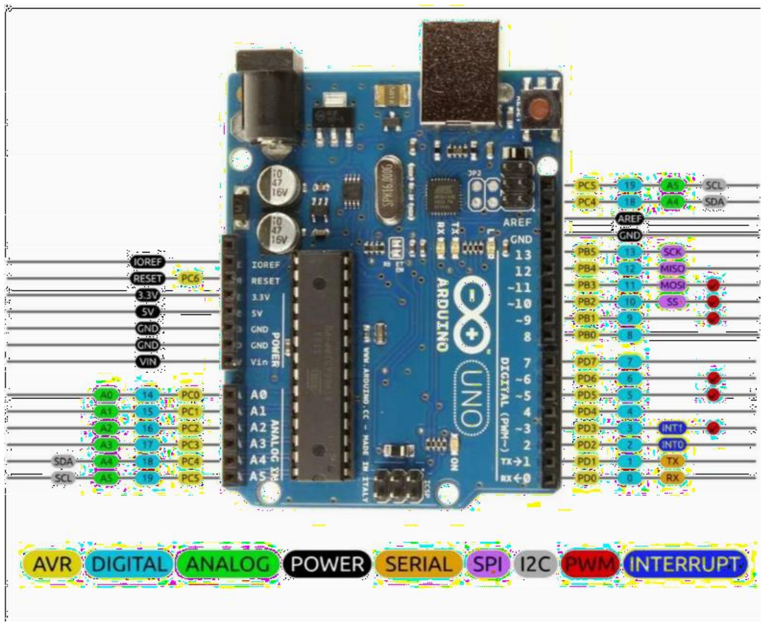

Fig No 4.2

- The Arduino UNO has a total of 28 pins, organized into different categories:
- Digital Pins (0-13): These pins can be used as input or output for digital signals. Pins 0 (RX) and 1 (TX) are used for serial communication.
- Analog Pins (A0-A5): These pins are used for analog input, capable of reading signals from analog sensors and converting them to a digital value.
- Power Pins:
  - o VIN: Input voltage to the Arduino when using an external power source (712V).
  - o 5V: Regulated 5V output from the onboard regulator.
  - o 3.3V: A 3.3V supply generated by the onboard regulator.
- GND: Ground pins.

- PWM Pins (~3, ~5, ~6, ~9, ~10, ~11): Digital pins that can output a PWM signal, useful for controlling devices like servos and LEDs.
- Reset: Can be used to reset the microcontroller.
- AREF: Reference voltage for the analog inputs.

4.1.3 Ease of use:

User-Friendly IDE: The Arduino IDE (Integrated Development Environment) is simple and intuitive, making it easy to write and upload code to the board.
Pluggable Modules: The board supports "shields," which are pluggable modules that add functionality without complex wiring.
Extensive Libraries: A wide range of libraries simplifies the process of interfacing with sensors, motors, and other peripherals.

4.1.4 Core Chip: ATmega328P

Architecture: The ATmega328P is an 8-bit AVR RISC-based microcontroller.

Clock Speed: It operates at a clock speed of 16 MHz.

Memory: It has 32 KB of flash memory for storing code, 2 KB of SRAM for variables, and 1 KB of EEPROM for data storage.

Operating Voltage: The chip operates at 5V but can handle a range of input voltages (7-12V) through the onboard voltage regulator.

4.1.5 Powerfulness of the Microcontroller:

- Processing Capabilities: The ATmega328P is powerful enough to handle a wide range of tasks, from simple digital control to complex sensor data processing.
- Flexibility: With its general-purpose I/O pins and support for various communication protocols (SPI, I2C, UART), the Arduino UNO can interface with a vast array of components and modules.
- Expandability: The ability to add shields and use libraries makes the Arduino UNO adaptable to numerous applications, from basic automation to sophisticated robotics.
- Community and Resources: The strength of the Arduino platform lies not only in the hardware but also in the extensive support from its community, providing ample resources, tutorials, and pre-built code libraries.

4.2 Temperature and humidity sensor

Temperature and humidity sensor (or rh temp sensor) is devices that can convert temperature and humidity into electrical signals that can easily measure temperature and humidity. Temperature humidity transmitters on the market generally measure the amount of temperature and relative humidity in the air, and convert it into electrical signals or other signal forms according to certain rules and output the device to the instrument or software to meet the environmental monitoring needs of users.

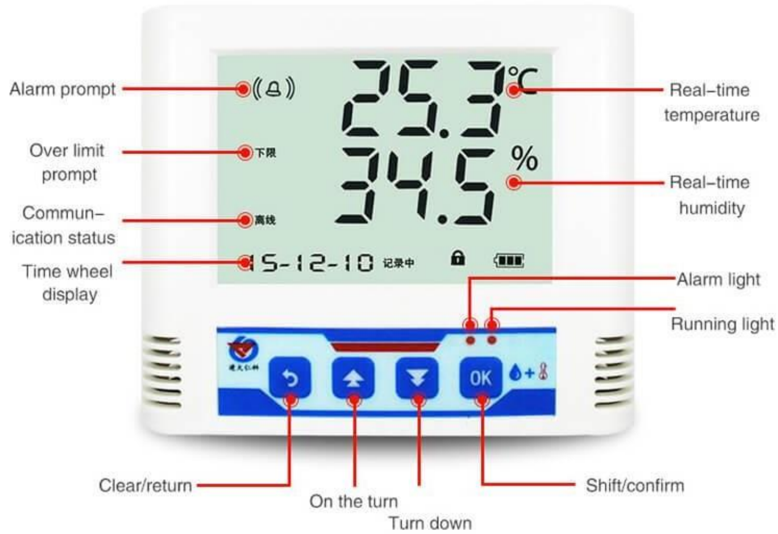

Fig No 4.3

Temperature and humidity sensors measure various substances based on changes in their electrical characteristics when temperature and humidity change. Depending on the type of sensor, the principle of operation varies. The operating principles of several common temperature and humidity sensors are described below.

4.2.1 Resistive temperature and humidity sensor working principle

The resistive temperature and humidity sensor measure the temperature and humidity in the air using an internal temperature-sensitive resistor (RTD) and a moisture-sensitive element resistance. The resistance value of this detection material changes with temperature and humidity and is converted into a corresponding electrical signal that outputs a temperature and humidity value.

4.2.2 Capacitive temperature and humidity sensor working principle

The capacitive temperature and humidity sensor working principle that the dielectric constant of a material changes with temperature and humidity. When the dielectric constant of the material changes, the inductive capacitance will also change accordingly, thus outputting the corresponding temperature and humidity values.

4.2.3 Electrothermal temperature and humidity sensor working principle

The electrothermal temperature and humidity sensor measure temperature and humidity by utilizing the heating and cooling process of an electric heating element. And the value of temperature and humidity is obtained by measuring the change in resistance value. They usually consist of a heating element, a temperature sensor and a humidity sensor.

4.2.4 Principle of humidity transducer

The components of the temperature and humidity sensor module mainly include a humidity sensitive capacitor and a conversion circuit. The humidity sensitive capacitor is composed of a glass substrate, a lower electrode, a humidity sensitive material, and an upper electrode.

Humidity sensitive material is a kind of high molecular polymer, its dielectric constant changes with the relative humidity of the environment. When the environmental humidity changes, the capacitance of the humidity sensitive element changes accordingly. That is, when the relative humidity increases, the humidity sensitive capacitance also increases, and vice versa. The conversion circuit of the sensor converts the change in humidity-sensitive capacitance into a change in voltage, which corresponds to a change in relative humidity of 0 to 100% RH. The output of the sensor shows a linear change of 0 to 1v.

- Best WiFi Temperature Humidity Sensor For Monitor Remotely
- Best Temperature And Humidity Data Loggers Introduce, Types And Select

Fig No 4.4

4.3 Rain Sensor

A **rain sensor** or rain switch is a switching device activated by rainfall. There are two main applications for rain sensors. The first is a water conservation device connected to an automatic irrigation system that causes the system to shut down in the event of rainfall. The second is a device used to protect the interior of an automobile from rain and to support the automatic mode of windscreen wipers.

4.3.1 Principle of operation

The rain sensor works on the principle of total internal reflection. An infrared light shone at a 45-degree angle on a clear area of the windshield is reflected and is sensed by the sensor inside the car. When it rains, the wet glass causes the light to scatter and a lesser amount of light gets reflected back to the sensor.

An additional application in professional satellite communications antennas is to trigger a rain blower on the aperture of the antenna feed, to remove water droplets from the mylar cover that keeps pressurized and dry air inside the wave-guides.

4.3.2 Irrigation sensors

Rain sensors for irrigation systems are available in both wireless and hard-wired versions, most employing hygroscopic disks that swell in the presence of rain and shrink back down again as they dry out — an electrical switch is in turn depressed or released by the hygroscopic disk stack, and the rate of drying is typically adjusted by controlling the ventilation reaching the stack. However, some electrical type sensors are also marketed that use tipping bucket or conductance type probes to measure rainfall. Wireless and wired versions both use similar mechanisms to temporarily suspend watering by the irrigation controller specifically they are connected to the irrigation controller's sensor terminals, or are installed in series with the solenoid valve common circuit such that they prevent the opening of any valves when rain has been sensed.

Some irrigation rain sensors also contain a freeze sensor to keep the system from operating in freezing temperatures, particularly where irrigation systems are still used over the winter.Some type of sensor is required on new lawn sprinkler systems in Florida, New Jersey, Minnesota, Connecticut and most parts of Texas.

4.3.3 Automotive sensors

Fig No 4.5

In 1958, the Cadillac Motor Car Division of General Motors experimented with a water-sensitive switch that triggered various electric motors to close the convertible top and raise the open windows of a specially-built Eldorado Biarritz model, in case of rain. The first such device appears to have been used for that same purpose in a concept vehicle designated Le Sabre and built around 1950–51.

General Motors' automatic rain sensor for convertible tops was available as a dealer-installed option during the 1950s for vehicles such as the Chevrolet Bel Air. For the 1996 model year, Cadillac once again equipped cars with an automatic rain sensor; this time to automatically trigger the windshield wipers and adjust their speed to conditions as necessary.

In December 2017 Tesla started rolling out an OTA update (2017.52.3) enabling their AP2.x cars to utilize the onboard cameras to passively detect rain without the use of a dedicated sensor.

Most vehicles with this feature have an auto position on the control column.

4.3.4 Physics of rain sensor

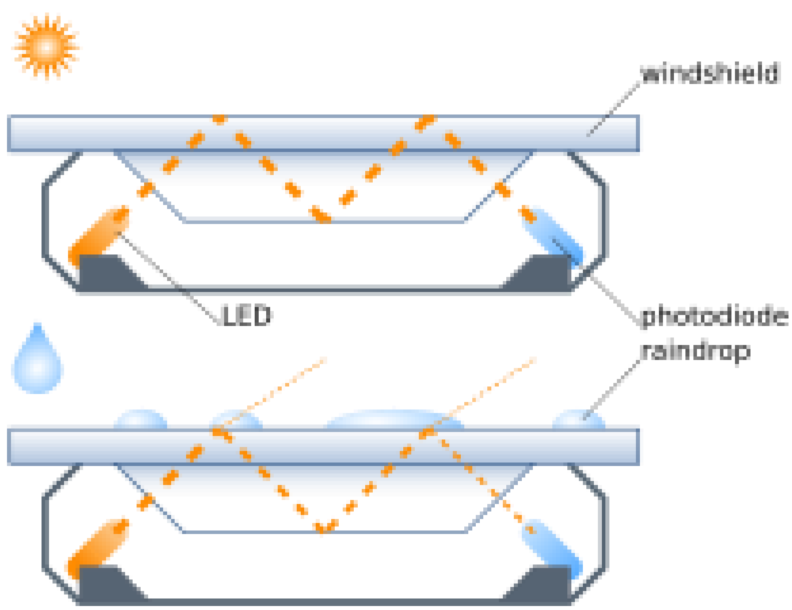

Fig No 4.6

The most common modern rain sensors are based on the principle of total internal reflection. At all times, an infrared light is beamed at a 45-degree angle into the windshield from the interior. If the glass is dry, the critical angle for total internal refraction is around 42°. This value is obtained with the total internal refraction formula

where is the approximate value on air's refraction index for infrared and is the approximate value of the glass refraction index, also for infrared.[5] In that case, since the incident angle of light is 45°, all the light is reflected and the detector receives maximum intensity.

If the glass is wet, the critical angle changes to around 60° because the refraction index of water is higher than air (In that case, because the incident angle is 45°, total internal reflection is not obtained. Part of the light beam is transmitted through the glass and the intensity measured for reflection is lower the system detects water and the wipers turn on.

4.4 Power Supply

A **power supply** is an electrical device that supplies electric power to an electrical load. The main purpose of a power supply is to convert electric current from a source to the correct voltage, current, and frequency to power the load. As a result, power supplies are sometimes referred to as electric power converters. Some power supplies are separate standalone pieces of equipment, while others are built into the load appliances that they power. Examples of the latter include power supplies found in desktop computers and consumer electronics devices. Other functions that power supplies may perform include limiting the current drawn by the load to safe levels, shutting off the current in the event of an electrical fault, power conditioning to prevent electronic noise or voltage surges on the input from reaching the load, power-factor correction, and storing energy so it can continue to power the load in the event of a temporary interruption in the source power.

All power supplies have a power input connection, which receives energy in the form of electric current from a source, and one or more power output or **power rail** connections that deliver current to the load. The source power may come from the electric power grid, such as an electrical outlet, energy storage devices such as batteries or fuel cells, generators or alternators, solar power converters, or another power supply. The input and output are usually hardwired circuit connections, though some power supplies employ wireless energy transfer to power their loads without wired connections. Some power supplies have other types of inputs and outputs as well, for functions such as external monitoring and control.

4.4.1 AC adapter

Fig No 4.7

An AC adapter is a power supply built into an AC mains power plug. AC adapters are also known by various other names such as "plug pack" or "plug-in adapter", or by slang terms such as "wall wart". AC adapters typically have a single AC or DC output that is conveyed over a hardwired cable to a connector, but some adapters have multiple outputs that may be conveyed over one or more cables. "Universal" AC adapters have interchangeable input connectors to accommodate different AC mains voltages.

Adapters with AC outputs may consist only of a passive transformer; in case of DC-output, adapters consist of either transformer with few diodes and capacitors or they may employ switch-mode power supply circuitry. AC adapters consume power (and produce electric and magnetic fields) even when not connected to a load; for this reason they are sometimes known as "electricity vampires", and may be plugged into power strips to allow them to be conveniently turned on and off.

4.5 Motor Driver

A **Motor driver**, also known as a **control motor**, is an electronic device or module that controls and manages the operation of an electric motor. It serves as an interface between a microcontroller or other control system and the motor itself, enabling precise control of the motor's speed, direction, and other parameters. Motor drivers are commonly used in various applications, including robotics, automation, automotive systems, and industrial machinery.This module comprises two servo and two DC motor drivers. The motors are devoid of + and – terminals, thereby eliminating the need to be concerned about reverse connections. The direction of the motors changes based on the wiring.

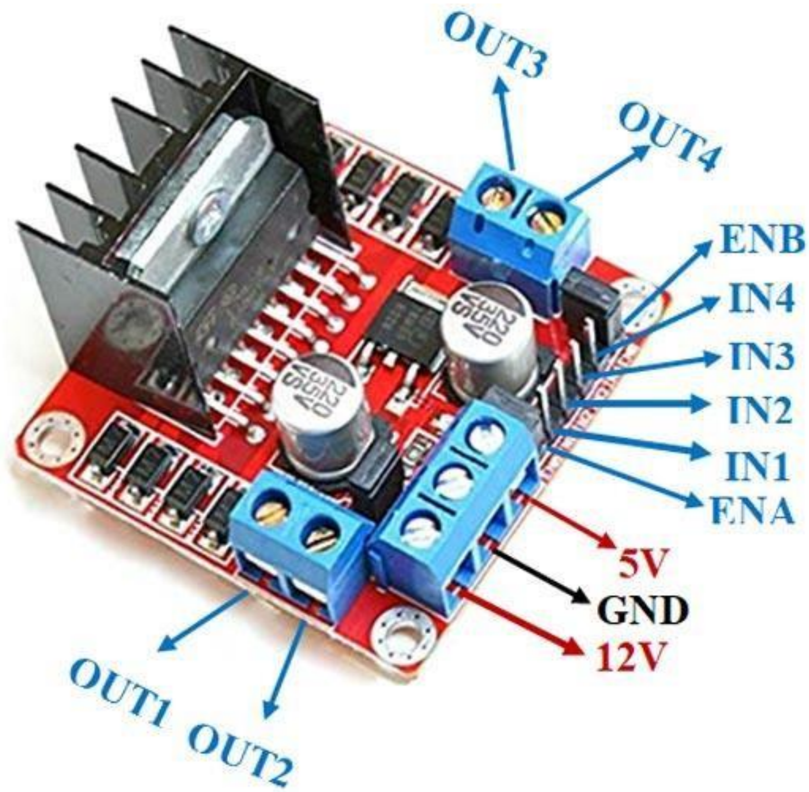

Fig No 4.8

4.6 Motor

A motor is an electrical machine that converts electrical energy into mechanical energy. It uses electromagnetic induction to produce rotation or linear motion, which can be used to power a wide range of devices and applications.

- Electrical energy is supplied to the motor through a power source.
- The electrical energy is converted into a magnetic field by the motor's windings.
- The magnetic field interacts with a rotor (a moving part) to produce rotation or linear motion.
- The rotation or linear motion is then transferred to a load, such as a wheel or a gear, to perform useful work.

Fig No 4.9

4.7 Roof

A roof is the uppermost part of a building or structure that provides protection from the elements, such as rain, snow, sunlight, and wind. It is a critical component of a building's envelope, serving as a barrier between the interior and exterior environments.

A roof typically consists of several layers, including:

- Roofing material (e.g., shingles, tiles, metal)
- Underlayment (e.g., felt paper, synthetic underlayment)
- Roof deck (e.g., plywood, oriented strand board)
- Support structure (e.g., rafters, trusses)

Fig No 4.10

The primary functions of a roof are:

- Weather protection
- Structural support
- Insulation and energy efficiency
- Aesthetics and architectural design

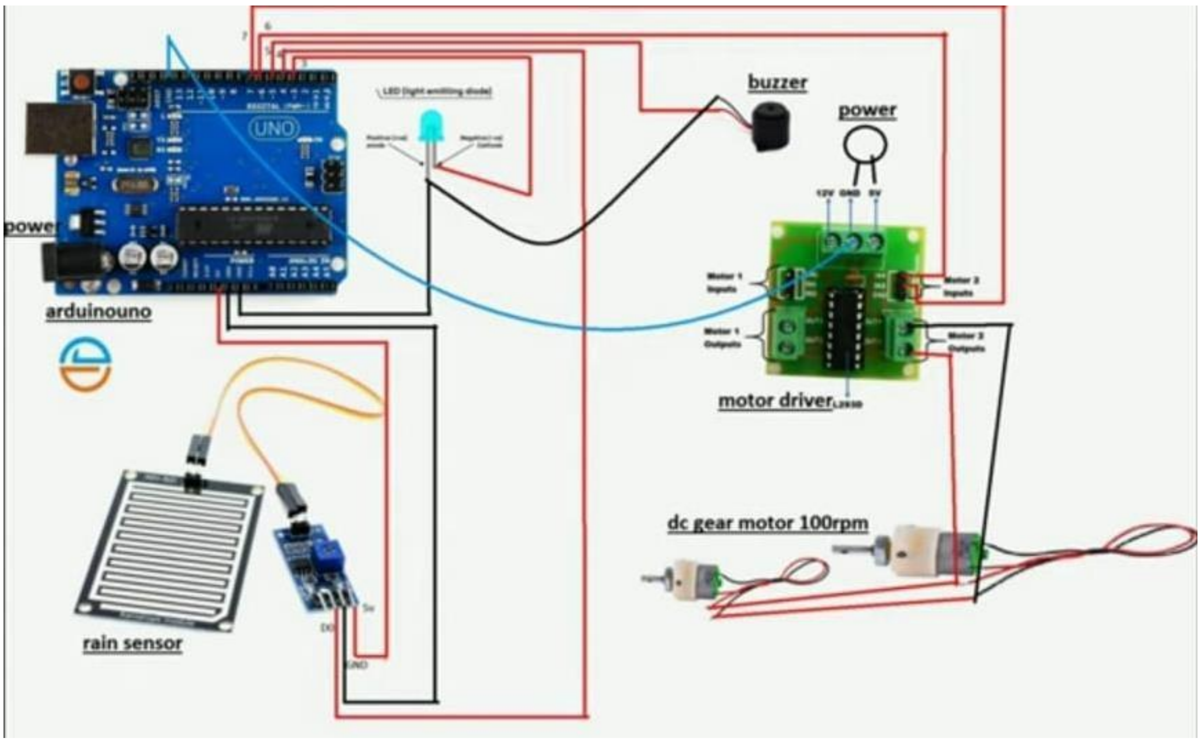

Fig No 4.11

Code:-

```
#define RAIN_SENSOR_PIN 2
#define MOTOR_A_IN1 3
#define MOTOR_A_IN2 4
#define MOTOR_B_IN3 5
#define MOTOR_B_IN4 6
enum RoofState { OPEN, CLOSED, MOVING_TO_CLOSE, MOVING_TO_OPEN };
RoofState roofState = OPEN; // Initial roof state
```

```
const unsigned long ROOF_MOVEMENT_TIME = 5000; // Simulated time to fully
open/close the roof in milliseconds
unsigned long movementStartTime = 0;
void setup() {
pinMode(RAIN_SENSOR_PIN, INPUT);
pinMode(MOTOR_A_IN1, OUTPUT);
pinMode(MOTOR_A_IN2, OUTPUT);
pinMode(MOTOR_B_IN3, OUTPUT);
pinMode(MOTOR_B_IN4, OUTPUT);
  Serial.begin(9600);
}
void loop() {
  int rainDetected = digitalRead(RAIN_SENSOR_PIN);
  Serial.print("Rain detected: ");
  Serial.println(rainDetected);
  unsigned long currentTime = millis();
  switch (roofState) {
    case OPEN:
      if (rainDetected == HIGH) {
        // Start closing the roof
        roofState = MOVING_TO_CLOSE;
        movementStartTime = currentTime;
        closeRoof();
      }
      break;
    case CLOSED:
      if (rainDetected == LOW) {
        // Start opening the roof
        roofState = MOVING_TO_OPEN;
        movementStartTime = currentTime;
        openRoof();
```

```
      }
      break;
    case MOVING_TO_CLOSE:
      if (currentTime - movementStartTime >= ROOF_MOVEMENT_TIME) {
        // Simulate roof fully closed
        stopMotors();
        roofState = CLOSED;
        Serial.println("Roof is now closed.");
      }
      break;
    case MOVING_TO_OPEN:
      if (currentTime - movementStartTime >= ROOF_MOVEMENT_TIME) {
        // Simulate roof fully opened
        stopMotors();
        roofState = OPEN;
        Serial.println("Roof is now open.");
      }
      break;
  }
  delay(100); // Short delay for stability
}
void closeRoof() {
  Serial.println("Closing roof...");
  digitalWrite(MOTOR_A_IN1, HIGH);
  digitalWrite(MOTOR_A_IN2, LOW);
  digitalWrite(MOTOR_B_IN3, HIGH);
  digitalWrite(MOTOR_B_IN4, LOW);
}
void openRoof() {
  Serial.println("Opening roof...");
  digitalWrite(MOTOR_A_IN1, LOW);
```

```
  digitalWrite(MOTOR_A_IN2, HIGH);
  digitalWrite(MOTOR_B_IN3,  LOW);
  digitalWrite(MOTOR_B_IN4, HIGH);
}
void stopMotors() {
  Serial.println("Stopping motors...");
  digitalWrite(MOTOR_A_IN1, LOW);
  digitalWrite(MOTOR_A_IN2, LOW);
  digitalWrite(MOTOR_B_IN3, LOW);
  digitalWrite(MOTOR_B_IN4, LOW);
}
```

5. Result

5.1 Primary Expected Results

- Effective Protection of Clothes: The automatic rain shelter should provide effective protection of clothes from rain, ensuring that they remain dry and clean.
- Convenient and Reliable Operation: The automatic rain shelter should be convenient and reliable to operate, with minimal user intervention required.
- Improved User Experience: The automatic rain shelter should improve the user experience, providing a comfortable and dry environment for users to enjoy the outdoors.

5.2 Secondary Expected Results

- Reduced Risk of Water Damage: The automatic rain shelter should reduce the risk of water damage to clothes, minimizing the need for costly repairs or replacements.
- Increased Durability of Clothes: The automatic rain shelter should increase the durability of clothes, by protecting them from the elements and reducing wear and tear.
- Enhanced User Safety: The automatic rain shelter should enhance user safety, by providing a secure and stable environment for users to seek shelter from the rain.

5.3 Tertiary Expected Results

- Reduced Maintenance Requirements: The automatic rain shelter should reduce maintenance requirements, by minimizing the need for manual intervention and reducing the risk of damage from the elements.

- Increased Energy Efficiency: The automatic rain shelter should increase energy efficiency, by using advanced materials and designs to minimize energy consumption.
- Improved Aesthetics: The automatic rain shelter should improve aesthetics, by providing a sleek and modern design that complements its surroundings.

5.4 Quantitative Expected Results

- Waterproofing Efficiency: The automatic rain shelter should achieve a waterproofing efficiency of at least 95%.
- Deployment Time: The automatic rain shelter should deploy in less than 30 seconds.
- Energy Consumption: The automatic rain shelter should consume less than 10 watts of power during operation.

Fig No 5.1 Model Presentation

6. Future Scope

6.1 Short-Term Future Scope (2025-2030)

- Integration with Wearable Technology: Integration of automatic rain shelter with wearable technology, such as smartwatches and fitness trackers, to provide users with real-time weather updates and alerts.
- Expansion to New Markets: Expansion of automatic rain shelter to new markets, such as Asia and South America, where there is a high demand for innovative rain protection solutions.
- Development of New Materials: Development of new materials and technologies, such as nanotechnology and advanced polymers, to improve the durability and water resistance of automatic rain shelter.
- Improvement of Deployment Mechanism: Improvement of the deployment mechanism of automatic rain shelter to make it faster, smoother, and more reliable.

6.2 Mid-Term Future Scope (2030-2035)

Integration with Internet of Things (IoT): Integration of automatic rain shelter with IoT devices and sensors to provide users with real-time weather updates and alerts.

- Development of Smart Rain Shelter: Development of smart rain shelter that can adjust its size and shape according to the user's needs and the weather conditions.
- Expansion to New Applications: Expansion of automatic rain shelter to new applications, such as outdoor furniture, bicycles, and motorcycles.
- Development of Sustainable Materials: Development of sustainable materials and technologies, such as bioplastics and recycled materials, to reduce the environmental impact of automatic rain shelter.

6.3 Long-Term Future Scope (2035-2040)

- Development of Autonomous Rain Shelter: Development of autonomous rain shelter that can deploy and adjust itself automatically without human intervention.
- Integration with Artificial Intelligence (AI): Integration of automatic rain shelter with AI algorithms to provide users with personalized weather forecasts and recommendations.

- Expansion to Space Exploration: Expansion of automatic rain shelter to space exploration, where it can provide protection for astronauts and spacecraft from harsh weather conditions.
- Development of Advanced Materials: Development of advanced materials and technologies, such as metamaterials and nanotechnology, to improve the performance and durability of automatic rain shelter.

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